

Finite Connected Externalities in Converter-Rich Power-System Dynamics: Re-Equilibrated Reconstruction, Pole Anatomy, and the Boundary of Modal Materiality

Authors withheld for double-blind review

Abstract—Engineering actions in converter-rich grids are commonly screened one at a time or through local sensitivities. Such calculations do not isolate the finite non-additive effect of a portfolio after its operating point has moved. This paper formulates a connected intervention calculus for finite actions on an index-one differential–algebraic power-system model. Every coalition is solved at its own ac equilibrium, reinitialized, and rebuilt before a band-integrated relative log-determinant is evaluated. Pair and triple Möbius externalities are reconstructed independently from connected total derivatives over the action hypercube. In a matched IEEE 39-bus benchmark with three grid-following converters, 266 of 672 pair cases and 147 of 1,344 triple cases exceeded five conservative uncertainty budgets. All 129 locked-holdout reconstructions closed within the same five-budget criterion. An exact Schur factorization then separates algebraic, mass-scaling, and finite-pole terms; its largest closure error is 7.39×10^{-13} . For the leading dispatch–network pair, a frozen-affine calculation underestimates the finite effect by approximately $6.3\times$ at the OP00 reference point. Yet the strongest externalities are concentrated in well-damped converter-control modes, whereas the actual limiting mode is a 1.22-Hz synchronous-electromechanical mode. Across all 15 retained coalitions, the largest baseline critical-mode damping externality is only 5.15×10^{-6} . Thus a reproducible dynamic externality can exist without materially consuming the registered damping margin. This distinction defines both the contribution and the limit of the method: it diagnoses finite interaction structure, but is not by itself a stability or mitigation certificate.

Index Terms—converter-driven stability, differential–algebraic equations, eigenvalue analysis, grid-following converters, higher-order interactions, Möbius inversion, power-system dynamics, spectral determinant

I. INTRODUCTION

CONVERTER-RICH power systems couple controller parameters, active-power schedules, and network conditions through both the ac operating point and the dynamic Jacobian. The effect of changing two devices together therefore need not equal the sum of changing them separately. This elementary observation is operationally important: screening tools are frequently local or one-action-at-a-time, while actual planning and tuning decisions arrive as finite portfolios.

Several mature theories already illuminate parts of this problem. Classical modal analysis relates states, residues, participations, and eigenvalue sensitivities to oscillatory behavior [1], [2]. Augmented and multi-parameter sensitivities can retain equilibrium dependence and higher-order terms [3],

[4]. Return-ratio and impedance methods expose feedback interactions without requiring the same internal coordinate description [5]–[7]. Recent work also uses open-loop modal proximity and second-order eigenvalue displacement to identify converter interaction modes [8].

These methods answer valuable but different questions. A derivative is local to its expansion point. An impedance or return-ratio criterion evaluates a feedback interconnection. A participation factor associates a mode with states. None of these objects, by itself, is the finite inclusion–exclusion remainder of a specified engineering portfolio after every subset has been physically applied and independently re-equilibrated. The combinatorial operation needed to isolate that remainder is Möbius inversion [9], [10]. The missing bridge is to apply it to a re-equilibrated differential–algebraic operator and then connect the finite result to derivative, factor, and modal descriptions without confusing existence with consequence.

This paper develops and tests that bridge. A normalized action coordinate maps zero to a registered baseline and one to a finite engineering change. For each coalition vertex, the ac power flow is solved again, dynamic states are initialized again, and the descriptor operator is rebuilt. A relative real log-determinant, integrated over a fixed frequency band, summarizes the resulting operator. Direct pair and triple Möbius differences define the finite externality. Independently, a connected set-partition formula integrates mixed total derivatives over the action hypercube. The two routes must close within a case-specific uncertainty budget.

The contribution is not a new Möbius identity, Schur complement, trace formula, perturbation determinant, or eigenvalue sensitivity. It is the following integrated, experimentally gated workflow:

- 1) a finite set function whose every vertex is an independently re-equilibrated power-system DAE;
- 2) a connected reconstruction that retains mixed physical and equilibrium curvature rather than only singleton feedback;
- 3) an exact descriptor-factor anatomy separating algebraic, mass-scaling, and finite-pole externalities;
- 4) a registered modal-materiality ladder that asks whether the externality reaches the mode that actually sets the damping margin.

Table I makes the novelty perimeter explicit. The first four rows are established approaches; none is displaced here.

Möbius inversion supplies the set-function coefficient, higher-order sensitivity supplies local derivative information, and return-ratio/impedance methods supply loop-level stability information. The proposed contribution is the registered combination of a finite re-equilibrated power-system set function, complete pair/triple subset evaluation, connected reconstruction, descriptor-factor anatomy, and a separate materiality test.

The tests use a canonical IEEE 39-bus system with three grid-following (GFL) converters and eight moderate actions. The main positive result is numerical and structural: reproducible pair and genuine third-order finite-pole externalities exist, and the connected reconstruction closes the direct finite computation. The main negative result is equally important: the largest externalities found in the registered search do not materially alter the limiting damping mode. A post-mortem identifies a family mismatch. The externality-dominant modes are well-damped converter-control or phase-locked-loop (PLL) modes, while the limiting mode is a lightly damped synchronous-electromechanical mode. Hence *dynamic externality existence and operational interpretation under a registered modal-materiality screen are distinct claims*.

This distinction narrows the practical interpretation. The calculus can tell an analyst that a finite portfolio contains reproducible, non-additive dynamic structure, quantify what a frozen model misses, and locate pole motion when a contour certificate is valid. It does not certify instability, identify an industrial mitigation, establish cycle or strongly connected component (SCC) causality, or transfer a mechanism to electromagnetic-transient (EMT) simulation. The cross-domain EMT work in this study validates only the matched benchmark infrastructure used to launch the phasor-domain campaign.

The remainder of the paper is organized as follows. Section II defines the finite calculus and exact factorization. Section III describes the registered benchmark, actions, uncertainty budgets, and holdout discipline. Section IV reports finite externalities and connected reconstruction. Section V separates algebraic and pole contributions and presents contour localization. Section VI reports the modal consequence and materiality boundary. Sections VII and VIII discuss interpretation, limitations, and conclusions.

II. FINITE CONNECTED INTERVENTION CALCULUS

A. Actions and Re-Equilibrated Descriptor Operator

Let $a = (a_1, \dots, a_m) \in [0, 1]^m$ denote normalized engineering actions. Coordinate $a_i = 0$ is the registered baseline and $a_i = 1$ is the complete finite action. Intermediate values provide a smooth interpolation; they are not a first-order approximation. At every a , the ac equations are solved and the dynamic model is reinitialized. Linearization of the resulting differential-algebraic equations gives

$$T(s, a) = \begin{bmatrix} sM(a) - f_x(a) & -f_y(a) \\ -g_x(a) & -g_y(a) \end{bmatrix}, \quad (1)$$

where x and y are differential and algebraic variables, respectively. All derivatives in this paper are *total* derivatives along the re-equilibrated action path. They therefore contain direct

parameter changes, the implicit movement of the equilibrium, and the resulting Jacobian curvature.

For a registered band $\Omega = [\omega_\ell, \omega_h]$ and fixed real contour shift σ , define

$$\Phi_\Omega(a) = \int_\Omega w(\omega) [\log |\det T(\sigma + j\omega, a)| - \log |\det T(\sigma + j\omega, 0)|] d\omega. \quad (2)$$

We use $w(\omega) = 1/[\omega \log(\omega_h/\omega_\ell)]$, so (2) is an average on logarithmic frequency. The baseline subtraction makes $\Phi_\Omega(0) = 0$. The real log-absolute determinant has no complex-log branch choice. Fixed nonsingular left and right matrix scalings add only action-independent constants and hence cancel in every higher-order finite difference.

Equivalently, define the finite-dimensional relative determinant $D(s, a) = \det T(s, a) / \det T(s, 0)$ wherever the fixed contour is regular; then the integrand in (2) is $\log |D(s, a)|$. This notation is only a finite non-normal determinant ratio. Contour statements below follow from the argument principle and registered numerical certificates, not from self-adjoint spectral-shift theory.

The log determinant is used as a compact characteristic functional, not as a standalone stability index. It aggregates how the full descriptor operator changes across the selected frequency band. Its factor and modal interpretations are tested later rather than assumed.

B. Direct Finite Externality

For a nonempty action set $S \subseteq \{1, \dots, m\}$, let 1_U activate exactly the subset $U \subseteq S$. The finite Möbius externality is

$$\Delta_S \Phi = \sum_{U \subseteq S} (-1)^{|S| - |U|} \Phi(1_U). \quad (3)$$

For a pair, this is the joint response minus both singleton responses plus the baseline. For a triple, it removes the three singleton and three pair contributions. A nonzero triple coefficient is thus a genuine third-order set-function dividend (Möbius coefficient) in the chosen finite set function. It is not, without additional evidence, an irreducible three-device causal mechanism.

Equation (3) is coordinate dependent in the ordinary sense that the physical definition and amplitude of each action matter. It supports an existential statement for the frozen actions, not a universal statement that all portfolios or parameterizations are non-additive.

C. Finite-to-Connected Bridge

If the re-equilibrated map is sufficiently smooth on the full $|S|$ -dimensional hypercube, repeated application of the fundamental theorem of calculus yields

$$\Delta_S \Phi = \int_{[0, 1]^{|S|}} \partial_S \Phi(a_S; a_{\bar{S}} = 0) da_S. \quad (4)$$

This identity turns a vertex-level Möbius coefficient into an integral of a connected mixed derivative. It requires a smooth valid path, not merely valid endpoints.

TABLE I
METHOD-TO-METHOD NOVELTY PERIMETER

Approach	Amplitude	Equilibrium treatment	Interaction object	Primary interpretation
First-order eigen-sensitivity	Local	One expansion point	Singleton derivative of a selected eigenvalue	Directional modal change
Higher-order total sensitivity	Local Taylor series	May include implicit equilibrium derivatives	Selected mixed derivatives to a truncation order	Local curvature and parameter ranking
Return-ratio / impedance	Frequency-domain model	Operating point supplied to the model	Ports, loops, or interconnection blocks	Nyquist/loop stability and coupling
Proximal eigenvalue displacement [8]	First-/second-order modal displacement	Not an all-subset finite-action requirement	Proximal open-loop modal coupling	Identification of converter interaction modes
Möbius set-function coefficient	Finite	Domain-agnostic vertex values	All subsets of a specified set	Pure higher-order set-function dividend
This work	Finite registered actions	New AC solution and DAE at every required vertex	All vertices of every registered pair and triple	Externality, exact factor anatomy, then separate registered modal materiality

Write $G = T^{-1}$ and let T_B be the mixed total derivative with labels in the nonempty block $B \subseteq S$. Jacobi's identity and repeated product differentiation give the complex connected derivative

$$\partial_S \log \det T = \sum_{\pi \in \text{Part}(S)} \frac{(-1)^{k-1}}{k} \times \sum_{\rho \in \text{Perm}(\pi)} \text{tr} \left(\prod_{r=1}^k GT_{B_{\rho(r)}} \right), \quad (5)$$

where $\pi = \{B_1, \dots, B_k\}$ is a set partition of S . The $1/k$ factor removes cyclic overcounting under the trace. The real part of (5), integrated over frequency and the action hypercube, reconstructs (3). For a pair,

$$\partial_{12} \log \det T = \text{tr}(GT_{12}) - \text{tr}(GT_1 GT_2), \quad (6)$$

which explicitly separates mixed curvature from the familiar feedback product. Triples contain intrinsic T_{123} , mixed-curvature/feedback, and three-factor connected feedback terms. Omitting T_{12} or higher mixed terms is a frozen-affine diagnostic, not the finite calculus.

D. Exact Descriptor-Factor Anatomy

Assume the algebraic Jacobian $g_y(a)$ and the registered differential mass block $M(a)$ are nonsingular. The model is then index one on the tested path, and the Schur identity gives

$$\det T(s, a) = \det[-g_y(a)] \det[sM(a) - A_r(a)], \quad (7)$$

$$A_r(a) = f_x(a) - f_y(a)g_y(a)^{-1}g_x(a), \quad (8)$$

$$\det[sM - A_r] = \det M \det[sI - A(a)], \quad A = M^{-1}A_r. \quad (9)$$

Consequently, the finite externality of (2) separates exactly:

$$\Delta_S \Phi = \Delta_S \Phi_{\text{alg}} + \Delta_S \Phi_{\text{mass}} + \Delta_S \Phi_{\text{pole}}. \quad (10)$$

The first term comes from $\det(-g_y)$, the second from $\det M$, and the third from $\det(sI - A)$. Algebraic and mass factors are frequency independent but may still depend on the action and contribute to the integrated finite functional. They vanish from an s derivative, not necessarily from (10).

The pole term permits a finite-dimensional perturbation-determinant reading, conceptually related to trace and determinant theories [11], [12]. We do not transfer self-adjoint

spectral-shift theorems to the non-normal power-system operator. Instead, pole attribution is accepted only when a registered complex contour avoids poles, preserves occupancy, and satisfies numerical quadrature and moment closure. Condition numbers and biorthogonal matching are retained because eigenvectors of a non-normal matrix can be sensitive [13].

III. REGISTERED BENCHMARK AND PROTOCOL

A. Matched IEEE 39-Bus Testbed

The principal case is a reconciled IEEE 39-bus system with GFL converter plants at buses 36, 37, and 38. The population remains GFL-only throughout the campaign; no GFL-to-grid-forming model substitution is part of the action space. The ANDES implementation uses a symbolic-numeric DAE framework [14]. Its static data were reconciled to a matched ParaEMT model before the main campaign.

The E02C prerequisite verified the common post-conversion operating point, network-wide stationarity, converter active/reactive power, current and PLL channels, adjacent-timestep convergence, and generalized-QZ eigenpair residuals. All seven registered readiness gates passed after retaining the canonical 0.97143 transformer tap. This establishes that the benchmark is suitable for the ANDES experiments below. It does *not* provide EMT validation of any externality, modal mechanism, or intervention.

The 100-MVA ac solve uses bus 39 as its reference/slack bus. GFL buses 36–38 are fixed- P/Q injections: their legacy PV voltage equations are disabled and their registered reactive injections are fixed, while the remaining synchronous PV controls and the slack close the network equations. Thus the A7 active-power increment is balanced by the reference generator through the power flow. Every accepted vertex was outside the implemented converter current and controller-saturation limits. Across the recorded E03 vertices, minimum bus voltage was 0.84056 pu and maximum GFL current was 0.90842 pu on the device base, below the 1.2-pu current limit.

B. Finite Engineering Actions

Table II lists the eight actions frozen before discovery. The controller actions preserve model class and state dimension. For PLL bandwidth actions, the proportional and integral gains follow $K_p = K_{p0}r$ and $K_i = K_{i0}r^2$ with

TABLE II
FROZEN FINITE ACTION SET

ID	Family	Location	Full action
A1	PLL bandwidth	bus 36	+20% natural frequency
A2	PLL bandwidth	bus 37	+20% natural frequency
A3	PLL bandwidth	bus 38	+20% natural frequency
A4	Current-command BW	bus 36	+20% bandwidth
A5	Current-command BW	bus 37	+20% bandwidth
A6	Current-command BW	bus 38	+20% bandwidth
A7	Active-power reference	bus 37	540 → 560 MW
A8	Network reactance	line 25–37	0.0232 → 0.02552 pu

$r = 1 + 0.2a_i$. The current-command bandwidth actions use the corresponding registered REGCP interpolation. A7 changes the GFL37 active-power reference by 20 MW (0.20 on the 100-MVA system base, 0.02 on the 1000-MVA device base), and A8 increases the series reactance of branch 25–37 by 10%. The three GFL ratings are 1000 MVA, with baseline active injections of 560, 540, and 830 MW. The frozen PLL natural frequency changes from 2.0 to 2.4 Hz; the current-command time constant changes from 0.0200 to 0.0167 s; and the branch-25–37 reactance changes from 0.0232 to 0.02552 pu. A8 was chosen by a static network rule: branch 25–37 is the median-reactance member of the three transformer branches incident to the GFL cluster. No dynamic result informed the choice.

The 24 preregistered operating points comprise 16 discovery and eight locked holdout seeds. Each point evaluates all 28 pairs and 56 triples, for 672 pair and 1,344 triple cases. A failed ac solution, initialization, descriptor build, or smooth-path check invalidates the case and remains in the failure ledger; cases are never silently dropped.

C. Spectral Functional and Numerical Uncertainty

The primary band is 0.1–30 Hz. Frequency integration uses Gauss–Legendre quadrature in log frequency with 48 nodes and a 96-node reference. A conditioning-only pilot selected the first admissible fixed contour shift, $\sigma = 0$, from a preregistered list. A 0.1–50 Hz band is retained only as a sensitivity analysis. Connected integration uses the frozen action-hypercube rule, complete pair/triple partitions, and sparse active-column trace solves; the implementation never forms the full T^{-1} for the campaign.

The direct uncertainty ϵ_{direct} is a conservative sum of case-specific frequency refinement, sparse-log-determinant, and power-flow/ Jacobian terms. A direct coefficient is called resolved only if

$$|\Delta_S \Phi| > 5\epsilon_{\text{direct}}. \quad (11)$$

The reconstruction criterion is

$$|\Delta_S \Phi_{\text{conn}} - \Delta_S \Phi_{\text{direct}}| \leq 5\epsilon_{\text{combined}}, \quad (12)$$

where the combined budget also includes derivative, action-grid, hypercube, and connected-frequency errors. Correlation is descriptive and cannot pass this gate.

The sequence of freezes separates method selection, discovery, and holdout. Numerical pilots may select only predeclared

TABLE III
REGISTERED EVIDENCE LADDER AND TERMINAL STATUS

ID	Question	Status
C1	Do finite re-equilibrated pair/triple externalities exist?	Supported
C2	Does connected reconstruction close the direct result?	Supported
D1	Does exact Schur factor anatomy close?	Supported
D2	Are finite-pole externalities reproducible?	Supported
D3	Can certified contours localize pole motion?	Supported, conditional
N1	Is nominal modal materiality established?	Rejected
N2	Is load-stress materiality established?	Rejected
N3	Is weak-grid materiality established?	Unresolved; baseline ineligible
N4	Is a cycle/SCC intervention established?	Unassessed

resolution parameters. Direct discovery determines which effects exceed their uncertainty. The connected implementation then reconstructs every resolved case. Finally, action coalitions are locked without inspecting modal consequence and evaluated on independent holdout seeds.

D. Modal Consequence and Stress Ladder

For an eigenvalue $\lambda = \alpha + j\beta$, modal frequency and damping are $f = |\beta|/(2\pi)$ and $\zeta = -\alpha/|\lambda|$. Left and right eigenvectors are biorthogonally normalized. Tracking uses a registered biorthogonal MAC (BMAC), eigenvalue distance, one-to-one assignment, and adaptive homotopy. A modal family must be locked in discovery before the holdout is viewed.

The finite damping externality $\Delta_S \zeta$ is computed with the same inclusion–exclusion rule as (3). The registered materiality threshold is $|\Delta_S \zeta| \geq 0.0025$ at four or more of eight holdout points, with at least six valid tracks and same-sign fraction at least 0.75. This 0.0025 value and the separate 5% minimum-damping screen are study criteria, not universal grid-code limits.

Two increasingly targeted stress tests were frozen after the nominal materiality claim failed. E05C scales load to the first 0.85-pu voltage boundary using eight new seeds and all 15 retained coalitions. E05D was intended to vary grid strength, but its immutable rule required a baseline minimum damping of at least 5% before any coalition could be evaluated. A later descriptive audit at $\kappa_{\text{grid}} = 1$ was explicitly barred from changing any gate.

E. Evidence Ladder

Table III distinguishes structural evidence from operational evidence. This ordering prevents an exact mathematical decomposition from being promoted automatically into a stability or mitigation claim.

IV. FINITE EXTERNALITIES AND CONNECTED RECONSTRUCTION

A. Direct Pair and Triple Effects

All 2,016 E03 cases were smooth-valid. Among 672 pair cases, 266 exceeded the five-budget resolution criterion;

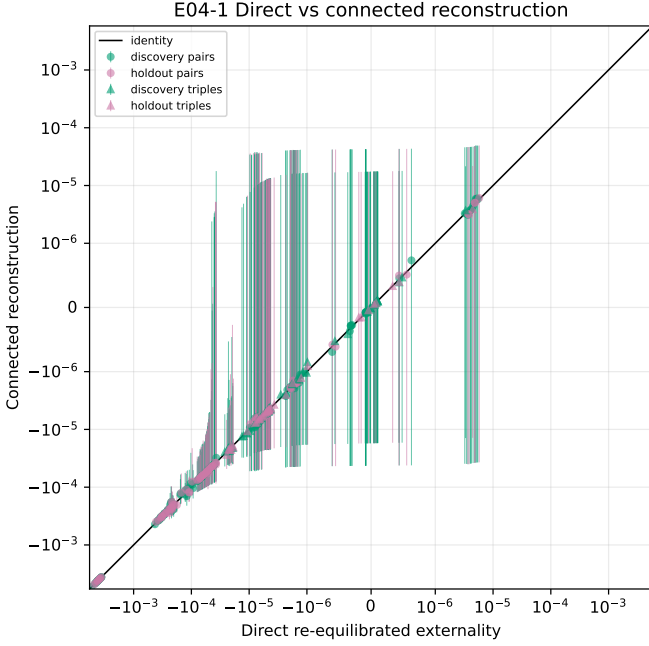


Fig. 1. Direct re-equilibrated externality versus connected reconstruction for every resolved discovery and holdout case. Error bars are conservative combined uncertainty; the signed symlog axes retain small and negative values. All 129 holdout cases close within five budgets.

among 1,344 triple cases, 147 did so. The unresolved majority, especially 1,197 of the triple cases, is part of the result: higher-order effects are not ubiquitous at the frozen amplitudes.

The strongest resolved discovery pair was A7–A8 at OP12, $\Delta_{\{7,8\}}\Phi = -0.00535233$ with direct uncertainty 6.93×10^{-6} . The largest locked-holdout value for the same pair was -0.00483156 . It combines a 20-MW dispatch increase at GFL37 with a 10% reactance increase on its incident branch. The leading triple was A2–A7–A8, which adds the bus-37 PLL bandwidth action. Its largest discovery coefficient was -2.64008×10^{-5} , and its largest holdout magnitude was 2.45036×10^{-5} .

The holdout replication rule was met by ten pairs and five triples, each with same-sign fraction 1.0. (The broader C1 gate counts one additional pair under its original population rule; the later E05B population contains the ten fully retained pairs used for anatomy and stress.) These are existential results on the frozen coordinates. They do not imply that every finite controller or network portfolio is non-additive.

B. Independent Connected Closure

All 413 effects resolved in E03 were reconstructed through (4)–(5). On the locked holdout, all 129 cases satisfied (12). Median absolute errors were 1.11638×10^{-6} for pairs and 5.80806×10^{-8} for triples. The overall mean bias, 2.45587×10^{-7} , was below the median combined uncertainty. Figure 1 shows direct and reconstructed values with case-specific uncertainty bars; the pointwise gate, rather than the apparent correlation, supports the closure claim.

This agreement checks more than a final scalar. The direct route uses only coalition vertices and Möbius differencing.

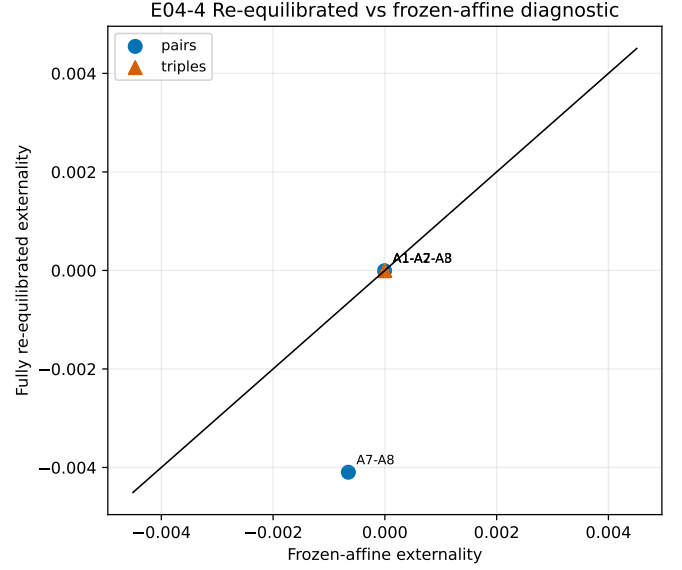


Fig. 2. Fully re-equilibrated versus frozen-affine externality at OP00 for registered examples. The dispatch–network pair A7–A8 lies far from the identity line: re-equilibration increases its magnitude by about $6.3\times$.

The connected route evaluates total matrix derivatives, the complete set-partition expansion, frequency integration, and action-hypercube quadrature. Closing the two independently implemented paths is therefore evidence for the calculus and its numerical implementation. It is not a causal identification theorem: both calculations operate on the same model and action definitions.

C. Why Re-Equilibration Is Material to the Calculation

The connected anatomy distinguishes intrinsic mixed derivatives, products of single-action feedback terms, and mixed curvature. For A7–A8 on holdout, the median curvature contribution was -0.00338136 and the median feedback contribution was -0.000694529 , giving a median singleton-feedback adequacy ratio of only 0.169. Thus most of this finite pair is associated with the operating-point/network curvature that appears when dispatch and branch reactance are changed together.

At OP00, the fully re-equilibrated A7–A8 externality is approximately -0.00411 , while the frozen-affine result is -0.00065321 . The frozen calculation therefore captures only about 15.9% of the finite magnitude, an underestimate of approximately $6.3\times$. Figure 2 shows the diagnostic for the registered exemplars. This factor belongs to A7–A8 at the reference operating point; it is not asserted as a universal correction factor.

The result sharpens the comparison with local sensitivity. Higher-order equilibrium sensitivities can be computed [3], [4]; the present calculus does not claim otherwise. Its distinction is that the target object is the exact finite coalition remainder, and that mixed curvature is retained along the entire finite action path rather than truncated at one expansion point.

For the resolved triples, the median adequacy of the pure-feedback description was 0.975323, but signed terms some-

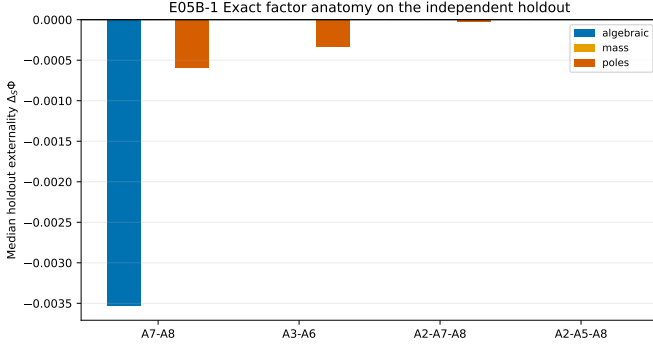


Fig. 3. Exact median holdout anatomy for registered coalitions. A7–A8 is dominated by the algebraic factor, while A3–A6 has zero connected algebraic/mass contribution and is pole dominated. Small triple coefficients are retained rather than visually inflated.

times canceled strongly. Accordingly, component signs and magnitudes are reported directly rather than converted into unstable percentage shares near a zero total. A large adequacy ratio for one action family does not justify discarding curvature for another.

V. EXACT FACTOR ANATOMY AND POLE LOCALIZATION

A. Schur Closure

The E05B holdout applied (10) to the 15 retained coalitions at 16 independent operating points. All builds were smooth-valid. The largest pointwise log-determinant factorization residual was 7.38964×10^{-13} ; the largest residual after frequency integration was 2.27374×10^{-13} ; and the largest residual after finite Möbius differencing was 3.41061×10^{-13} . These values are consistent with floating-point roundoff relative to the observed externalities.

The factorization answers a question that the aggregate characteristic functional cannot: is a finite externality produced predominantly by algebraic re-equilibration, action-dependent dynamic scaling, or pole motion? Figure 3 reports the median holdout anatomy for four registered examples.

For A7–A8, the median full externality was -0.00411289 . Its algebraic, mass, and pole components were -0.00353228 , approximately zero, and -0.00059089 , respectively. The leading spectral interaction is therefore largely an ac re-equilibration effect in the descriptor determinant, although it also contains a reproducible pole component.

A3–A6 provides the complementary case. These two co-located action families act on the bus-38 PLL and current-command bandwidths. Their connected algebraic and mass terms are zero at the recorded precision, while the median pole externality is -0.00033610 . This is a direct counterexample to the idea that every resolved log-determinant externality is merely an algebraic power-flow artifact. Ten pairs and five triples exhibit reproducible finite-pole externalities on the E05B holdout.

The triple results require careful language. A nonzero third-order coefficient in Φ_{pole} is a genuine third-order *finite-pole externality*: after removing all lower-order set contributions, the pole factor remains non-additive. It does not show that

three physical devices form an isolated causal loop. Such a statement would require a frozen localization object, a mechanism-specific intervention, and a matched negative control, none of which is present here.

B. Connected Logarithmic Derivative and Contour Moments

Differentiating only the pole factor with respect to s removes the frequency-independent algebraic and mass factors:

$$\frac{\partial}{\partial s} \log \det(sI - A) = \text{tr}[(sI - A)^{-1}]. \quad (13)$$

Applying the finite connected difference to (13) gives a resolvent-based description of non-additive pole displacement. Registered closed contours then provide zeroth- and first-order moments. A certificate is accepted only when the contour stays clear of poles at every required action vertex, occupancy is consistent, quadrature refinement closes, and direct and connected moments agree.

Three candidates met these conditions across all 16 holdout points: the pair A7–A8 and triples A2–A7–A8 and A2–A5–A8. For A7–A8, the median absolute first-moment shift was $4.74913 \times 10^{-3} \text{ s}^{-1}$; the median real and imaginary components were 4.65632×10^{-3} and $-8.99434 \times 10^{-4} \text{ s}^{-1}$, respectively. Maximum numerical contour error was 1.24×10^{-15} . Both triple candidates likewise retained valid one-pole certificates and same-sign fraction 1.0.

The A3–A6 contour did not pass. Its maximum numerical contour error was 0.9151, leaving zero valid-resolved holdout points under the registered rule. This negative case is retained. The factor decomposition still proves that A3–A6 has a pole externality; it does not authorize a contour-localized pole claim for that coalition.

C. Relation to Return-Ratio, Impedance, and Spectral Methods

Return-difference matrices and impedance Nyquist tests diagnose feedback stability and interconnection effects [5]–[7]. They can often be measured or assembled with fewer internal model details than (1). Proximal eigenvalue-displacement methods identify modes whose coupling invalidates a first-order open-loop approximation [8]. That recent comparator returns converter interaction modes from open-loop proximity and first-/second-order eigenvalue displacement. It overlaps with the present interest in modal interaction, but does not define a finite action-coalition Möbius estimand, require a new AC equilibrium at every subset vertex, separate descriptor-factor externalities, or test the independently identified minimum-damping family. The present method is neither its replacement nor a new stability criterion.

Its output is instead an attribution on a finite action set. The same physical portfolio is evaluated at every subset, and inclusion–exclusion isolates what belongs only to the joint finite action. The connected formula then states how that set-function coefficient is assembled from total operator derivatives. Finally, (10) asks whether the coefficient lies in algebraic re-equilibration, mass scaling, or finite poles. These questions are complementary to loop stability and modal proximity.

The determinant language also requires restraint. In finite non-normal matrices, log determinants and contour moments are computational identities under their stated regularity conditions. They do not inherit the ordering, positivity, or causal interpretation of self-adjoint spectral-shift theory. This study therefore uses *connected pole localization*, not spectral causality.

VI. FROM DYNAMIC EXTERNALITY TO MODAL MATERIALITY

A. Nominal Modal Consequence

The E05 modal study selected three coalitions mechanically from E03/E04 before examining any damping result: A7–A8 as the largest reproducible curvature-dominated pair, A3–A6 as the largest reproducible feedback-dominated pair, and A2–A7–A8 as the largest reproducible triple. Each candidate achieved valid mode tracking at all eight holdout points, with consistent signs.

The median finite damping externalities were -1.84336×10^{-4} for A7–A8, -4.09123×10^{-4} for A3–A6, and $+1.18769 \times 10^{-5}$ for A2–A7–A8. A7–A8 and A3–A6 also produced median total joint damping changes of -0.00391371 and -0.00565123 , respectively. Thus their portfolios change modal damping, but most of that change is additive at the tracked-mode level. None of the 24 candidate/seed points reached $|\Delta_S \zeta| \geq 0.0025$. The nominal materiality claim was therefore rejected under its preregistered rule.

This failure is not a contradiction. The characteristic functional integrates the complete descriptor over a frequency band, while a damping gate observes one tracked modal family. A large value of $\Delta_S \Phi$ may arise from algebraic re-equilibration, pole motion in a well-damped family, or a distributed collection of small shifts. Only the last stage asks whether the non-additive effect consumes a decision-relevant margin.

B. Load-Conditioned Stress

E05C tested whether modal composition amplified as load increased toward the first 0.85-pu voltage boundary. Eight new seeds, all 15 retained coalitions, and nine stress levels were evaluated with 100% valid externality and tracking rates. The feasible endpoint multiplier ranged from 1.05225 to 1.083125 across seeds.

Define the real-part composition ratio

$$\rho_{\text{comp}} = \frac{\Re\{\lambda_S - \lambda_S^{(<|S|)}\}}{\max(|\Re\{\lambda_S^{(<|S|)}\}|, \epsilon_\lambda)}, \quad (14)$$

where $\lambda_S^{(<|S|)}$ is the lower-order inclusion–exclusion prediction and ϵ_λ is the registered numerical floor. The largest observed ρ_{comp} was 0.00211758, and the largest candidate median at the final stress point was 0.00197909. No point reached the material damping threshold. There were no hard composition failures and no 5% screen crossings attributable to composition before the voltage boundary. Both load-conditioned amplification and material consequence were rejected.

Although ρ_{comp} was correlated with conditioning and remaining margin, those regressions were explanatory and did not change a candidate decision. A statistically monotone trend cannot substitute for the frozen magnitude gate.

C. Weak-Grid Attempt: Baseline Ineligibility

Conventional short-circuit ratio (SCR) is useful context for GFL operation in weak networks, but multi-infeed interactions can make single-POI strength metrics incomplete [15], [16]. Converter control can also introduce weak-grid dynamics that require more than static strength screening [17]. E05D therefore preregistered a grid-strength path with an immutable eligibility rule: the empty-coalition baseline had to retain at least 5% damping in the true minimum-damping 0.1–30 Hz mode.

At $\kappa_{\text{grid}} = 1$, all eight new baselines already violated that screen. Their minimum damping ranged from 0.0416536 to 0.0452751. The corresponding conventional single-POI values were $\text{SCR}_{36}=2.60275\text{--}2.64401$, $\text{SCR}_{37}=2.57783\text{--}2.65366$, and $\text{SCR}_{38}=1.32141\text{--}1.40561$. Because there was no eligible starting point, no safe stress endpoint existed and no E05D coalition outcome was inspected.

The only valid status is *unresolved–baseline ineligible*. It is neither a rejection nor support of weak-grid materiality. The 5% rule was not changed, the stress direction was not reversed, and no replacement candidate search was opened.

D. Critical-Mode Baseline Audit

After E05D closed, a separate descriptive audit asked why the eligibility screen and the externality search referred to such different margins. It used only $\kappa_{\text{grid}} = 1$, the eight E05D seeds, all 15 frozen E05B coalitions, and the original A1–A8 definitions. It did not create a new gate.

For each empty baseline, the audit selected the true minimum-damping positive-imaginary eigenvalue in 0.1–30 Hz, retained its left/right vectors, condition number, BMAC, and state and descriptor participations, and tracked that exact mode through every coalition subset. Global one-to-one matching required BMAC at least 0.90, with registered 5-, 9-, and 17-point homotopies. All 200 unique subset tracks and all 120 coalition/seed comparisons succeeded.

All eight limiting modes were synchronous-electromechanical. Their frequencies were 1.21973–1.22906 Hz, damping ratios were 0.0416536–0.0452751, and synchronous-machine/governor/exciter states carried 0.98619–0.98982 of normalized descriptor participation. By contrast, the 15 externality-dominant development modes were PLL/current-control families with damping 0.49390–0.99168. Figure 4 exposes both the damping and physical-family separation.

The limiting family is a coherent multi-machine electromechanical mode rather than a single converter-local mode. Median descriptor endogeneity is led by the rotor speed/angle channels of GENROU32, GENROU30, GENROU34, GENROU31, and GENROU33; the median angle-channel shares are 0.1450, 0.0849, 0.0816, 0.0727, and 0.0547, respectively.

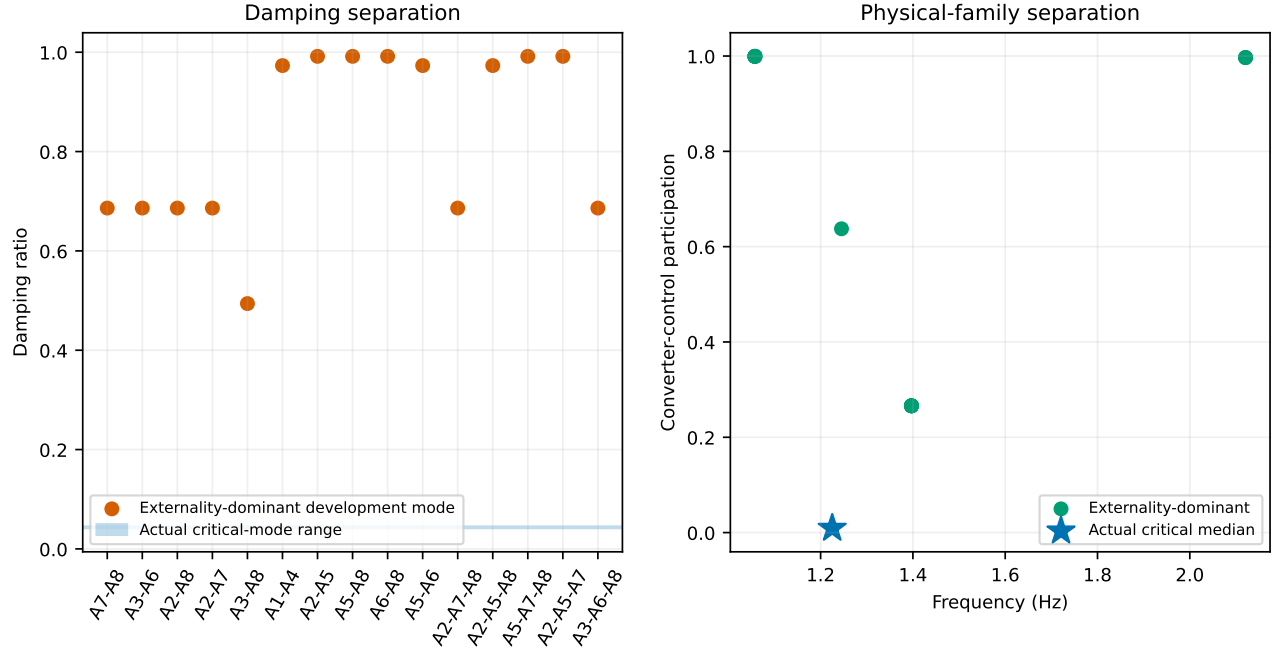


Fig. 4. Externality-dominant development modes versus the actual margin-setting mode at $\kappa_{\text{grid}} = 1$. Left: the development modes are much more strongly damped than the 4.17–4.53% critical-mode band. Right: they are converter-control/PLL dominated, whereas the critical mode is synchronous-electromechanical near 1.22 Hz.

TABLE IV
EXTERNALITY EXISTENCE VERSUS MODAL MATERIALITY

Evidence	Value	Interpretation
A7–A8 spectral externality	−0.00411 median holdout	Exists; algebraic dominated
A3–A6 pole externality	-3.36×10^{-4}	Exists; pole dominated
Nominal $ \Delta_S \zeta $	4.09×10^{-4} median	Below 0.0025 gate
Load-stress ρ_{comp}	0.00211758 maximum	No material point
Weak-grid campaign	8/8 baselines below 5%	Ineligible; unresolved
Critical-mode audit	5.15×10^{-6} max $ \Delta_S \zeta $	Below registered gate on limiting family

This state anatomy supports the family label without promoting it to a causal intervention target.

Across all 120 coalition/seed cases, the largest critical-mode $|\Delta_S \zeta|$ was 5.14674×10^{-6} . The maximum ρ_{comp} was 1.39286×10^{-4} for A7–A8 on seed D02, where $\Delta_{\{7,8\}} \Re \lambda = 4.56767 \times 10^{-5} \text{ s}^{-1}$. The most stabilizing ratio was -9.63916×10^{-6} for A5–A8, and the median absolute ratio over all cases was 6.38782×10^{-6} . These values are descriptive evidence for a mode-family mismatch, not a post hoc reinterpretation of E05D.

Table IV summarizes the resulting boundary. Structural externalities survive every exactness and reproducibility check, while operational materiality does not survive the registered consequence tests.

VII. DISCUSSION

A. What the Calculus Adds

The central practical object is the finite coalition remainder. Suppose an analyst plans to change a converter set point, retune a controller, and alter a nearby network element. Standard first-order screening estimates each increment around one baseline and then sums. The present calculation instead asks: after every subset has moved to its own ac equilibrium and its DAE has been rebuilt, what part of the full-portfolio characteristic change cannot be assigned to lower-order subsets?

That question is useful even when the answer is not operationally material. It distinguishes three failure modes of a simpler approximation:

- 1) *finite-action error*: a local Taylor model is inaccurate over the specified amplitude;
- 2) *re-equilibration error*: the operating point and algebraic Jacobian move jointly with the actions;
- 3) *interaction-order error*: pair or triple terms remain after all lower-order terms are removed.

The A7–A8 result contains all three. At OP00, a frozen-affine representation misses about 84.1% of the finite coefficient, while exact factor anatomy shows that its largest contribution is algebraic re-equilibration. A3–A6, however, has a reproducible pole-only coefficient. The method thus prevents the blanket conclusion that a characteristic externality must be merely a power-flow artifact or a dangerous control mode.

The connected reconstruction also supplies a stringent internal check. Direct Möbius differencing and integrated connected traces have different numerical error paths. Their

locked-holdout agreement makes it difficult for a sign convention, missing partition, or omitted curvature term to survive undetected. Exact Schur closure adds a second independent identity at the descriptor-factor level.

B. Finite Portfolio Information Absent from a Baseline First-Order Screen

A first eigenvalue sensitivity can rank actions for one mode near one equilibrium. Higher-order and equilibrium-point sensitivities extend that local picture [3], [4]. Selective modal analysis can reduce the calculation to relevant state groups [1]. These remain the appropriate tools when the decision question is local and the limiting modal family is known.

The finite calculus contributes when the decision variable is a named portfolio at a non-infinitesimal amplitude. It does not approximate the Möbius coefficient from a derivative at zero; it evaluates the coefficient directly and then reconstructs it by integrating the derivative over the whole hypercube. This is why mixed re-equilibration curvature matters. The 6.3-fold A7–A8 discrepancy would not be repaired merely by better tracking of the baseline eigenvalue.

Conversely, the critical-mode audit shows what the spectral functional can miss if used alone: relevance to the limiting mode. The functional found converter-control interactions reproducibly, and the pole factor verified that some were dynamic. But these modes were strongly damped. A separate minimum-damping search was required to show that the operative margin belonged to a synchronous family and that the externality barely moved it. Modal association tools—BMAC, participation, residues, condition numbers, and homotopy—support that family identification; none is a causal proof.

C. Power-System Relevance

Three features make the experiment more than a matrix identity exercise. First, actions are recognizable engineering changes: 20% controller-bandwidth adjustments, a 20-MW dispatch change, and a 10% branch-reactance change. Second, every coalition is required to satisfy the ac equations, dynamic initialization, and smooth-path checks. Third, the strongest pair couples dispatch and its incident network path, exactly the setting in which frozen network or fixed-operating-point screening is most vulnerable.

The negative materiality result is also relevant. Interaction-detection methods are often motivated by weak-grid risk, but interaction is not synonymous with limiting instability. In this benchmark, conventional SCR values provide useful context yet do not identify which modal family sets the 5% screen. The critical mode is synchronous-electromechanical despite the presence of three GFL plants and conspicuous converter-mode externalities. Planning studies should therefore distinguish at least three questions:

- 1) Is a portfolio non-additive in a chosen dynamic functional?
- 2) Does the non-additivity arise in algebraic coordinates or finite poles?
- 3) Does it materially change the mode or constraint that limits the operating decision?

The first two can be positive while the third is negative.

D. Limitations

The action space contains eight coordinates and only one matched IEEE 39-bus GFL testbed. The results establish reproducible existence, not prevalence across converter vendors, control architectures, larger networks, or action amplitudes. The three converter models share a common class. Grid-forming controls, hybrid populations, protection and saturation events, unbalanced dynamics, and topology-discontinuous actions remain outside scope.

The spectral functional is dimensionless and band dependent. Its sign and magnitude are not direct measures of damping, instability probability, energy, or economic cost. The normalized logarithmic frequency weight and 0.1–30 Hz band were frozen before the campaign, but another defensible functional may rank coalitions differently.

The finite-to-connected identity requires smooth re-equilibrated interpolation throughout the hypercube. It does not cover a path that crosses a power-flow bifurcation, limiter discontinuity, state-dimension change, or singular algebraic Jacobian. The exact Schur anatomy further assumes index-one behavior, nonsingular g_{yy} , and a nonsingular registered dynamic mass block. Contour localization is conditional on pole-free boundaries and numerical occupancy certificates; A3–A6 demonstrates that a pole externality may exist even when a selected contour cannot certify its localized moment.

The materiality ladder is deliberately narrow. The 0.0025 damping-externality threshold and 5% screen are preregistered study rules, not universal standards. E05C ended at a voltage boundary before finding material composition. E05D could not start because all empty baselines were already below the screen; it therefore leaves weak-grid materiality unresolved. No coalition result from E05D was viewed.

Finally, no cycle or SCC localization, mechanism-targeted correction, matched-cost negative control, nonlinear time-domain consequence, or blind EMT transfer was performed. E02C validates the benchmark infrastructure only. Consequently, this paper does not offer an industrial mitigation or claim that a particular interaction graph is causal.

E. Implications for Future Use

The terminal findings suggest a disciplined workflow for other systems when finite portfolio composition is the decision question. First, define finite actions in physical units and solve every coalition vertex. Second, resolve direct externalities against numerical uncertainty. Third, use connected reconstruction as an implementation and anatomy check. Fourth, split algebraic and pole factors exactly. Fifth, identify the actual constraint-setting modal family independently of the externality search. Only if the same family carries a material, reproducible finite effect should an intervention experiment be designed.

This ordering limits unnecessary controller surgery. In the present testbed, the descriptive audit explains why further tuning of the externality-dominant converter modes would not answer the registered stability-margin question. The correct terminal outcome is a boundary statement, not a new stress search.

VIII. CONCLUSION

This paper introduced a finite connected intervention calculus for converter-rich power-system DAEs. The defining choice is physical re-equilibration: each subset of a finite action portfolio receives its own ac solution, dynamic initialization, and descriptor operator. Direct Möbius differences isolate pair and triple externalities, and connected total derivatives reconstruct them over the full action hypercube.

The registered IEEE 39-bus campaign supports five bounded conclusions. Finite pair and genuine third-order externalities exist reproducibly. The connected reconstruction closes all locked-holdout cases within five conservative uncertainty budgets. Exact Schur anatomy separates algebraic, mass-scaling, and finite-pole contributions to roundoff. Reproducible finite-pole externalities exist, and certified contours localize pole motion for three retained coalitions. Re-equilibration can be quantitatively decisive: the frozen-affine approximation underestimates A7–A8 by about $6.3\times$ at the OP00 reference point.

The experiment also establishes a firm limit. None of the selected externalities met the preregistered nominal or load-conditioned damping materiality criteria. The weak-grid test was baseline-ineligible and remains unresolved. A separate audit showed why: the externality-dominant modes are well-damped converter-control families, whereas the actual limiting mode is a roughly 1.22-Hz synchronous-electromechanical family with only microunit-scale finite damping composition.

The resulting conclusion is deliberately narrower than an interaction-based stability claim: finite dynamic externalities can be real, higher order, and reproducible without materially consuming the system's registered modal margin. Detecting interaction structure is therefore a useful diagnostic step when finite portfolio composition is the decision question, but it is not a substitute for identifying the constraint-setting mode or for demonstrating a mechanism-specific intervention.

DATA AND CODE AVAILABILITY

The frozen protocols, manifests, source code, machine-readable tables, failure ledgers, and figure-generating inputs used in this paper are included with the review package. Each terminal experiment records source and input hashes. The E05D weak-grid package is retained unchanged, and the descriptive critical-mode audit is stored separately so that it cannot alter a registered gate.

ETHICS, CONTRIBUTIONS, FUNDING, AND COMPETING INTERESTS

No human participants, animals, or personal data were involved. Author names and CRediT roles are withheld for double-blind review and must be completed at submission. Funding and competing-interest declarations are likewise withheld during review and must be supplied by the authors before publication.

GENERATIVE-AI DISCLOSURE

A generative-AI coding assistant was used to help organize evidence, draft prose, and check software artifacts. All numerical claims were taken from frozen, executable experiment outputs; the authors are responsible for the analysis, verification, citations, and final text.

REFERENCES

- [1] I. J. Pérez-Arriaga, G. C. Verghese, and F. C. Schweppe, "Selective modal analysis with applications to electric power systems, part i: Heuristic introduction," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-101, no. 9, pp. 3117–3125, 1982.
- [2] F. L. Pagola, I. J. Pérez-Arriaga, and G. C. Verghese, "On sensitivities, residues and participations: Applications to oscillatory stability analysis and control," *IEEE Transactions on Power Systems*, vol. 4, no. 1, pp. 278–285, 1989.
- [3] H.-K. Nam, Y.-K. Kim, K.-S. Shim, and K. Y. Lee, "A new eigen-sensitivity theory of augmented matrix and its applications to power system stability analysis," *IEEE Transactions on Power Systems*, vol. 15, no. 1, pp. 363–369, 2000.
- [4] E. A. Zamora-Cárdenas and C. R. Fuerte-Esquivel, "Computation of multi-parameter sensitivities of equilibrium points in electric power systems," *Electric Power Systems Research*, vol. 96, pp. 246–254, 2013.
- [5] A. G. J. MacFarlane, "Return-difference and return-ratio matrices and their use in analysis and design of multivariable feedback control systems," *Proceedings of the Institution of Electrical Engineers*, vol. 117, no. 10, pp. 2037–2049, 1970.
- [6] J. Sun, "Impedance-based stability criterion for grid-connected inverters," *IEEE Transactions on Power Electronics*, vol. 26, no. 11, pp. 3075–3078, 2011.
- [7] A. Rygg, M. Molinas, C. Zhang, and X. Cai, "A modified sequence-domain impedance definition and its equivalence to the dq-domain impedance definition for the stability analysis of AC power electronic systems," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 4, no. 4, pp. 1383–1396, 2016.
- [8] L. K. Dewangan, F. G. Puricelli, and J. Beerten, "Interaction assessment of converter-interfaced power systems using proximal eigenvalue displacement approach," *IEEE Transactions on Power Systems*, vol. 41, no. 4, pp. 2454–2467, 2026.
- [9] G.-C. Rota, "On the foundations of combinatorial theory i. theory of möbius functions," *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, vol. 2, no. 4, pp. 340–368, 1964.
- [10] M. Grabisch, "k-order additive discrete fuzzy measures and their representation," *Fuzzy Sets and Systems*, vol. 92, no. 2, pp. 167–189, 1997.
- [11] M. G. Krein, "On the trace formula in perturbation theory," *Matematicheskii Sbornik (N.S.)*, vol. 33(75), no. 3, pp. 597–626, 1953. [Online]. Available: <https://www.mathnet.ru/eng/sm5397>
- [12] B. Simon, "Notes on infinite determinants of hilbert space operators," *Advances in Mathematics*, vol. 24, no. 3, pp. 244–273, 1977.
- [13] L. N. Trefethen, "Pseudospectra of linear operators," *SIAM Review*, vol. 39, no. 3, pp. 383–406, 1997.
- [14] H. Cui, F. Li, and K. Tomsovic, "Hybrid symbolic-numeric framework for power system modeling and analysis," *IEEE Transactions on Power Systems*, vol. 36, no. 2, pp. 1373–1384, 2021.
- [15] H. Xin, W. Dong, X. Yuan, D. Gan, K. Wang, and H. Xie, "Generalized short circuit ratio for multi power electronic based devices infeed to power systems," *Proceedings of the CSEE*, vol. 36, no. 22, pp. 6013–6027, 2016.
- [16] North American Electric Reliability Corporation, "Integrating inverter-based resources into low short circuit strength systems," NERC, Reliability Guideline, Dec. 2017.
- [17] L. Fan, "Modeling type-4 wind in weak grids," *IEEE Transactions on Sustainable Energy*, vol. 10, no. 2, pp. 853–864, 2019.